

1.What is Data Loading in Power BI and why is it considered the first step of analysis?

Answer:

Data Loading in Power BI is the process of importing data from differences such as Excel,CSV,SQL databases or online services into Power BI. It is considered as the first step of anaysis because data must be available in Power BI before performing cleaning,transformation,visualization or report creation.

Examble:

In the Corono_Virus_2020 datase,details like country,State,Confirmed Cases and Vaccination Status are first loaded ino Power BI for further analysis.

2.Explain the difference between “Load” and “Transform Data” in Power BI.

Answer:

Load:Load is the process of importing data directly into Power BI from sources like Excel,CSV or SQL databases.In this step,the data is added without making any changes.

Transform Data:Transform Data is the process of cleaning and modifying the data before loading it into Power BI.It includes removing duplicates,replacing null values ,changing data types and filtering data to improve data quality .

3.What is a Fact Table and a Dimension Table? Give examples from the dataset.

Answer:

Fact Table:A Fact table contains numerical or measurable values used in analysis.

Example:

- Confirmed Cases
- Recovered Cases
- Death Cases

Dimension Table: A Dimensional Table contains descriptive information that explains the facts.

Example:

- Country
- State
- Date
- Vaccination Status

Fact and Dimension tables together help in creating proper data models in Power BI.

4. Why is Star Schema preferred over Snowflake Schema in Power BI?

Answer:

Star Schema is preferred because it is simple, easy to understand and provides better performance in Power BI. In this schema, one Fact Table is directly connected to multiple Dimension Tables.

Advantages:

- Faster query performance.
- Easy data modelling.
- Better report generation.
- Requires fewer table joins.

Compared to Snowflake Schema, Star Schema reduces complexity and improves analysis speed.

5. Identify and remove duplicate records based on Date, Country, and State?

Answer:

- Transform the data .
- Open Power Query Editor.

- Select Date, Country and State Columns.
- Click on **Remove Duplicates** under the Home tab.

After clicking it, Power BI removes all repeated records from the selected rows.

The screenshot shows the Power Query Editor interface. The main area displays a table with the following columns: Country, State, Date, Confirmed_Cases, Recovered_Cases, Deaths, and Active_Case. The table contains 32 rows of data. The 'APPLIED STEPS' pane on the right shows the following steps: Source, Promoted Headers, Changed Type, and **Removed Duplicates** (which is highlighted). The 'PROPERTIES' pane on the right shows the name 'coronavirus_2020_dataset'.

	Country	State	Date	Confirmed_Cases	Recovered_Cases	Deaths	Active_Case
1	Brazil	State_A	43831	41240	56133	158	
2	UK	State_B	43832	83807	61268	1977	
3	Italy	State_B	43833	33434	38243	956	
4	UK	State_A	43834	69724	57384	2071	
5	UK	State_C	43835	75697	31653	403	
6	USA	State_B	43836	28732	26557	4611	
7	Italy	State_B	43837	11314	39952	2779	
8	Italy	State_B	43838	23954	22180	1387	
9	Italy	State_A	43839	57108	78411	3453	
10	UK	State_D	43840	87044	1622	955	
11	Brazil	State_B	43841	87505	48756	1712	
12	Italy	State_D	43842	36668	40809	4703	
13	Spain	State_C	43843	23061	16725	1273	
14	UK	State_B	43844	79252	61396	3811	
15	USA	State_B	43845	39749	51226	2625	
16	Brazil	State_C	43846	16907	29402	2382	
17	Spain	State_B	43847	18777	4263	1441	
18	Spain	State_D	43848	93229	46828	2979	
19	USA	State_A	43849	51745	47924	1797	
20	Brazil	State_D	43850	51865	74182	4390	
21	UK	State_A	43851	81818	59052	4009	
22	India	State_A	43852	23624	11664	4850	
23	Brazil	State_D	43853	23321	35737	1394	
24	USA	State_D	43854	10488	29373	4281	
25	Spain	State_D	43855	22574	36397	1269	
26	UK	State_D	43856	5240	38709	4732	
27	Brazil	State_C	43857	43221	61120	3714	
28	India	State_B	43858	10478	19250	529	
29	India	State_A	43859	55645	21172	3623	
30	Italy	State_D	43860	19262	11555	2958	
31	Italy	State_D	43861	52796	78617	2914	
32							

6. Identify and replace null values in Vaccination_Status?

Answer:

- Open Power Query Editor.
- Select the Vaccination_Status Column.
- Use **Replace Values** option.
- Replace "null" values with "unkown" or "Not Vaccinated".

This helps in maintaining clean and complete data.

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Queries [1]

coronavirus_2020_dataset

Table.ReplaceValue(*Removed Duplicates*, "", "Unknown", Replacer.ReplaceValue, {"Vaccination_Status"})

	Id	Recovered_Cases	Deaths	Active_Cases	Tests_Conducted	Hospitalized	Vaccination_Status
1	41240	56133	158	41424	123788	10288	No
2	83807	61268	1977	44002	98239	17709	No
3	33434	38243	956	53744	27928	17543	Yes
4	69724	57384	2071	4795	281466	4650	Unknown
5	75697	31653	403	10634	451930	4987	Unknown
6	28732	26657	4611	4798	351040	15200	Yes
7	11314	39952	2779	22420	63292	4697	No
8	23954	22180	1387	51507	30055	16724	Unknown
9	57108	78411	3453	55039	306035	3828	No
10	87044	1622	955	29300	82912	18352	Yes
11	87505	48756	1712	24345	138660	17087	No
12	36668	40809	4703	10749	353680	12605	Unknown
13	23061	16725	1273	38595	269448	5482	Unknown
14	79252	61396	3811	25121	8345	18184	No
15	39749	51226	2625	37487	161372	5463	Yes
16	16907	29402	2362	28411	418475	13860	No
17	18777	4263	1441	33361	92006	4912	Yes
18	93229	46828	2979	21223	482769	4648	No
19	51745	47924	1797	7394	409568	855	Unknown
20	91865	74182	4390	14056	130350	2866	Yes
21	81818	59052	4009	28380	165437	11834	Yes
22	23624	11664	4850	26035	282590	16294	Yes
23	23321	35737	1394	30620	265480	16566	Unknown
24	10488	29373	4281	19877	325176	3112	No
25	22574	36397	1269	59945	207487	4492	Unknown
26	5240	38709	4732	55316	98852	15114	Yes
27	43221	61120	3714	43404	493092	1517	No
28	10478	19250	529	51824	139593	16018	Unknown
29	55645	21172	3623	53352	203779	17740	Yes
30	19262	11555	2958	30911	155553	11431	Yes
31	52796	78617	2914	22784	281618	8210	Unknown
32	...	...	...	...	...	...	...

10 COLUMNS, 500 ROWS Column profiling based on top 1000 rows

Query Settings

PROPERTIES

Name

coronavirus_2020_dataset

APPLIED STEPS

Source

Promoted Headers

Changed Type

Removed Duplicates

Replaced Value

PREVIEW DOWNLOADED AT 10:45

7.Create a new column to calculate Recovery Rate?

Answer:

- Open Table View.
- At the top menu **Table Tools** and add new column.
- Click on **New Column**
- Write the formula:
 - Recovery Rate = ('coronavirus_2020_dataset'[Recovered_Cases] / 'coronavirus_2020_dataset'[Confirmed_Cases] *100)
- Press Enter.

Now the Recovery Rate column will be created.

Table: coronavirus_2020_dataset (500 rows) Column: Recovery Rate (500 distinct values)

8. Create a summarized table showing total confirmed cases by Country.

Answer:

- Click on **Data View** icon
- Click Table Tools.
- Click on **New Table**
- A formula bar will appear at the top.
 - Country Summary =
`SUMMARIZE('coronavirus_2020_dataset','coronavirus_2020_dataset'[Country],"Total Confirmed Cases",SUM('coronavirus_2020_dataset'[Confirmed_Cases]))`
- Press Enter .

Power BI will automatically create a new summarized table called **Country Summary**.

